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Department of Electronics and Communication Engineering

Project Title:

Voice Recorder and Replay Circuit

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Internal Guide:

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Motivation/ Problem Statement

- Security systems
- Assistive Technology
- Educational devices
- Medical devices
- Military and aviation
- Event recorders
- Emergency systems
- In many cases and investigation, higher officials lose the case in the court because of no evidence of video or audio proof, if in case of audio proof alone Tamil nadu police solved over 100+ cases per year
- This incident motivates me to do this project.

Abstract

- APR 9301 is a single chip Voice recorder and Playback device for 20 to 30 seconds voice recording and play back. It is an ideal IC for automatic answering machine, door phones etc. This IC has data storage capacity and requires no software and microcontroller. It provides high quality voice recording and play back up to 30 seconds.

The IC requires minimum components to create a voice recorder. The IC has non volatile flash memory technology with 100K recording cycles and 100 year message retention capacity. The IC utilizes the Invox proprietary analog / multi level flash non volatile memory cells that can store more than 256 voltage levels. It requires a single 5 volt supply and operates in 25 mA current.

Literature Survey

Author Name	Paper Title	Journal Name	Vol No	Issue No & Page No
Shu Eang, Jiahao Cao, Xu He, Kun Sun, Qi Li	When the differences in the frequency domain are compensated	arXiv preprint	1	1 , 10-20 Published on September 1,2020
Yuan Gong, Jian Yang, Christian Poellabauer	Detecting the replay attacks using multi- channel audio	arXiv preprint	2	1 , 20-25 Published on March 18,2020
Jacob Huber, Mitchell Macknight, Christian Poellabauer	ReMASC: Realistic Replay attack corpus for voice controlled systems	arXiv preprint	3	2 , 50-56 Published on April 6,2019

Outcomes From Literature Survey

Paper Title	Observations (Methodology, Techniques Used, Advantages, Limitations & Result Discussion)
When the differences in the frequency domain are compensated	Methods – Signal analysis, Frequency compensation Techniques- Fourier Transform(FT), Inverse Fourier Transform(IFT), Windowing Functions, Equalisation. Advantages- Improved signal quality, better performance, flexibility, adaptability, Limitations- complexity, latency, overfitting, noise amplification, assumption dependence Result- Enhanced signal fidelity, reduced distortion, system efficiency, Real time feasibility.
Detecting the replay attacks using multi-channel audio	Methods – Data collection, Feature extraction, Replay Attack Detection. Techniques- Spatial Analysis, Time frequency analysis, Beamforming. Advantages- Increased accuracy, Automation. Limitations- Environmental Dependence, Complexity, Hardware requirements. Result- Detection accuracy, Enhancement security in biometric authentication.

Proposed Methodology

- The proposed methodology for a voice recorder and replay circuit involves using a microphone to capture voice signals, which are then amplified using a pre-amplifier circuit.
- The amplified analog signal is converted to digital form using an ADC (Analog-to-Digital Converter) and processed/stored in non-volatile memory such as EEPROM, Flash memory, or an SD card, controlled by a microcontroller (e.g., Arduino, STM32).
- For playback, the stored digital data is retrieved, converted back to an analog signal using a DAC (Digital-to-Analog Converter), and amplified for output through a speaker.
- This design ensures efficient voice recording, storage, and high-fidelity replay.

Software/ Hardware Requirements

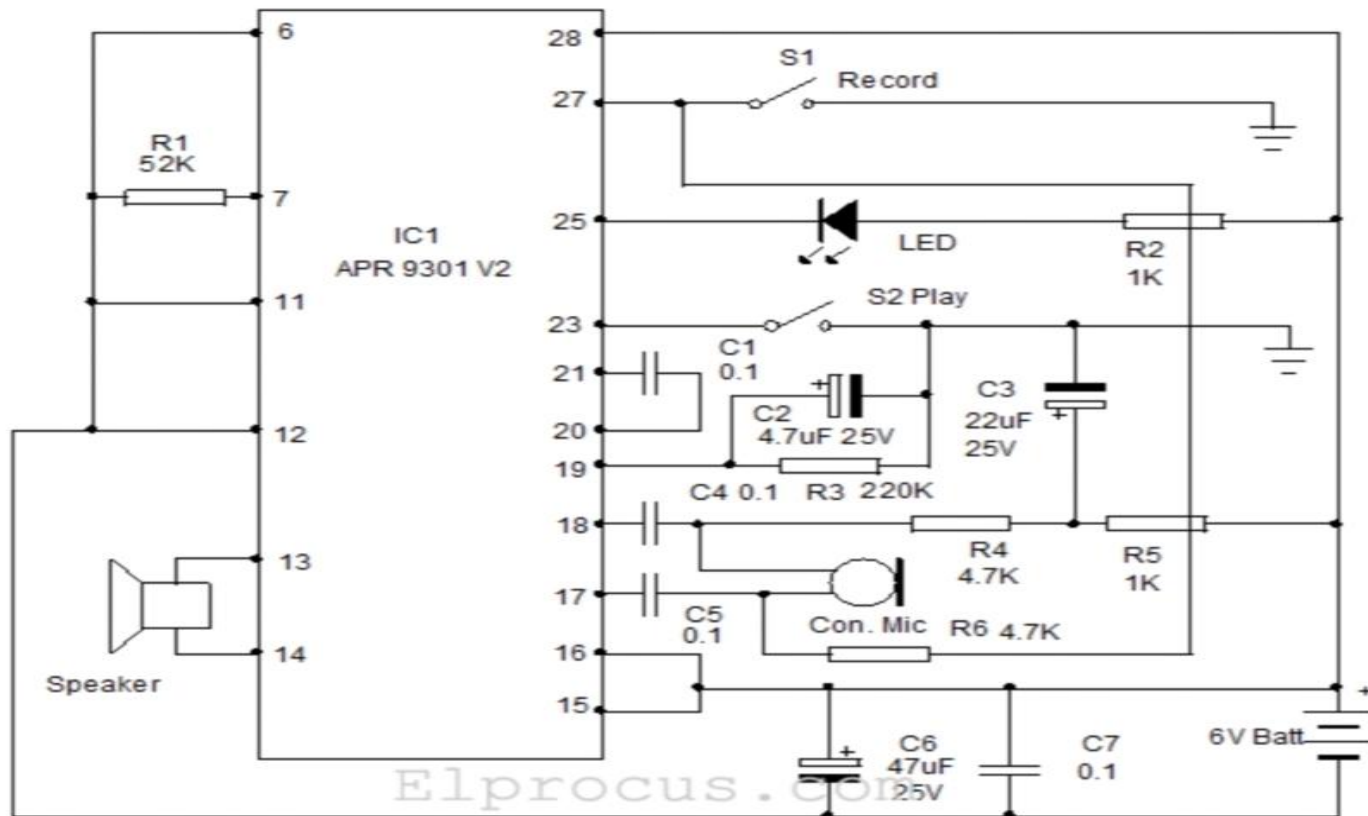
SOFTWARE REQUIREMENTS:

- Integrated Development Environment (Arduino IDE) ,
- Analog to analog converter (ADC) , Digital to analog converter (DAC)
- Audio processing libraries: (Audio compressing and filtration)
- Programming language: Python,

HARDWARE REQUIREMENTS:

- LED- 1.
- Speaker -1
- IC APR 9301 -1
- 1K Ohms – 2
- 4.7 Ohms – 2
- 52K – 1
- 220 Ohms – 1
- Capacitor 0.1 uF – 3
- Capacitor 22 uF – 1
- Capacitor 4.7 uF -1
- Capacitor 47 uF -1
- Push switch – 2
- Transformer 9V
- Power supply

Block Diagram/ Work Flow/ Flow Chart



Results

- The final results of the voice recorder and replay circuit demonstrate successful voice recording with minimal noise, achieved through an efficient microphone and pre-amplifier stage.
- The ADC accurately converts analog signals to digital format for reliable storage in non-volatile memory, ensuring data retention even after power loss.
- During playback, the DAC and amplifier stages provide clear and high-fidelity audio output.
- The circuit operates with low power consumption, making it suitable for portable applications, and uses cost-effective components, ensuring affordability.
- Overall, the compact design is practical for integration into various consumer devices.

Results



Image link:

<https://images.app.goo.gl/tnqq73DSniuFt1n78>

Reference

1. Books:

- Sedra, A. S., & Smith, K. C. (2015). Microelectronic Circuits. Oxford University Press.
- Boylestad, R. L., & Nashelsky, L. (2020). Electronic Devices and Circuit Theory. Pearson Education

2. Research Papers and Journals:

- Kaur, H., & Singh, S. (2019). “Design and Implementation of Voice Recording and Playback Systems.” International Journal of Electronics and Communication Engineering.
- Gupta, A., & Kumar, R. (2020). “Low-Power Embedded Voice Recorder for IoT Applications.” IEEE Access.

3. Web Resources:

- Arduino-based voice recorder tutorials (e.g., www.arduino.cc, www.instructables.com).
- Circuit design websites: www.electronicsforu.com, www.allaboutcircuits.com.

Thank You